

Publications

A. PhD dissertation

2005: “Statistical signal processing for spectrometry: application to pileup correction for Gamma spectrometry”, Department of Signal and Image Processing, Telecom ParisTech, Paris, France.

Advisors: Prof. Eric Moulines and Dr. Antoine Souloumiac (currently Prof. Antoine Souloumiac).

The dissertation was published in French and contains 160 pages. An appendix in the dissertation is written in English (30 pages) and summarizes the main results. Two journal papers were written based on the thesis: [26], [25].

The work was presented at several conferences: [13, 14,15].

Patents (Patents section: 2,3) were submitted on the digital implementation of the algorithm detailed in the dissertation.

B. Scientific Books (Refereed)

N/A

A. Authored Books – Published

N/A

B. Edited Books and Special Issues-Published

N/A

C. Other Scientific Publications

N/A

D. Journal Articles

Published:

- [1] D. Bykhovsky, Z. Chen, Y. Huang, X. Zheng and **T. Trigano**, ”Advanced Spectroscopy Time-Domain Signal Simulator for the Development of Machine and Deep Learning Algorithms”, to appear in *IEEE Sensors Letters*, 2025. (Q2, IF 2.2, GR h-5 34)
- [2] Y. Huang, X. Zheng, Y. Zhu, **T. Trigano**, Z. Chen and D. Bykhovsky, ”Deep learning based pile-up correction algorithm for spectrometric data under high count rate measurements”, *Sensors*, Special Issue on Spectral Detection Technology, Sensors and Instruments, Vol 25(5), 1464, Feb. 2025. (Q2, IF 3.4, GR h-5 191)

- [3] Z. Chen, D. Bykhovsky, X. Zheng, **T. Trigano**, Y. Zhu, "GaSim: A Python Class to Generate Simulated Time Signals for Gamma Spectroscopy", to appear in *SoftwareX*, 2025. (Q2, IF 2.4, GR h-5 45)
- [4] Y. Tzror, M. Bezner, S. Deri, **T. Trigano**, K. Ben-Harush, "Nanofilament organization in highly tough fibers based on lamin proteins", *Journal of the Mechanical Behavior of Biomedical Materials*, 160, 106748, 2024. (Q1, IF 3.3, GR h-5 83)
- [5] **T. Trigano** and D. Bykhovsky, "Deep Learning Based Method for Activity Estimation from Short-Duration Gamma Spectroscopy Recordings," *IEEE Transactions in Instrumentation and Measurement*, Aug. 2024. (Q1, IF 5.6, GR h-5 139)
- [6] **T. Trigano**, S. Talala, and D. Luengo, "Adaptive Trend Filtering for ECG Denoising and Delineation," *IEEE Journal of Biomedical and Health Informatics*, Vol 27(12), Dec. 2023, pp. 5755-5766. DOI: 10.1109/JBHI.2023.3314983 (Q1, IF 5.6, GR h-5 103)
- [7] **T. Trigano** and Z. Fradkin, "Fast Algorithm for Time Decay Estimation with Applications to Electrostatic Ion Beam Traps," *Measurement Science and Technology*, Vol 34(2), 2023, pp. 025701. DOI: 10.1088/1361-6501/ac9c23 (Q2, IF 2.7, GR h-5 156)
- [8] **T. Trigano** and D. Luengo, "Intracardiac ECG Pulse Localization using Overlapping Block Sparse Reconstruction", *Biological Signal Processing and Control*, Vol. 73(1), pp. 103921, Jan. 2023. DOI: <https://doi.org/10.1016/j.bspc.2022.103921> (Q1, IF 4.9, GR h-5 127) - 1 citation
- [9] M. Khayat, S. Deri, D. Wolf, **T. Trigano**, O. Medalia, K. Ben-Harush, "Biomimetic nuclear lamin fibers with remarkable toughness and stiffness", *International Journal of Biological Macromolecules*, Vol 163, pp 2060-2067, Nov. 2020. DOI: <https://doi.org/10.1016/j.ijbiomac.2020.09.113> (Q1, IF 7.7, GR h-5 213) - 4 citations.
- [10] **T. Trigano**, S. Vaknin and D. Luengo, "Fast Proximal Optimization for Sparse Reconstruction with Dictionaries Based on Translated Waveforms", *Signal Processing*, Vol 169(4), paper 107379, April 2020. DOI: <https://doi.org/10.1016/j.sigpro.2019.107379> (Q1, IF 3.4, GR h-5 112) - 4 citations.
- [11] **T. Trigano** and Y. Bechor, "Fast Background Removal of JPEG Images Based on HSV Polygonal Cuts for a Foot Scanner Device", *Journal of Real-Time Image Processing*, 17(4), pp. 981-992, 2019. DOI: <https://doi.org/10.1007/s11554-019-00850-5> (Q2, IF 2.9, GR h-5 43) - 4 citations.
- [12] D. Luengo, D. Meltzer and **T. Trigano**, "An Efficient Method to Learn Overcomplete Multi-Scale Dictionaries of ECG Signals", *Applied Science*, 8(12), 2018. DOI: <https://doi.org/10.3390/app8122569> (Q2, IF 2.7 GR h-5 165) - 12 citations.
- [13] **T. Trigano** and Y. Sepulcre, "Data-driven Parameter Selection in Activity Estimation for Nuclear Spectroscopy", *Signal Processing*, 151, 99-106, 2018. DOI: <https://doi.org/10.1016/j.sigpro.2018.05.006> (Q1, IF 3.4, GR h-5 112) - 3 citations.

- [14] D. Bykhovsky and **T. Trigano**, “Numerical Generation of Compound Random Processes with Arbitrary Autocorrelation Functions”, *Fluctuations and Noise Letters*, Nov. 2017. DOI: <https://doi.org/10.1142/S0219477518500013> (Q3, IF 1.2 GR h-5 34) - 3 citations.
- [15] **T. Trigano**, I. Shevtsov and D. Luengo, “CoSA : an Accelerated ISTA Algorithm for Dictionaries Based on Translated Waveforms”, *Signal Processing*, 139, 131-135, Oct. 2017. DOI: <https://doi.org/10.1016/j.sigpro.2017.04.004> (Q1, IF 3.4, GR h-5 112) - 14 citations.
- [16] **T. Trigano** and Y. Cohen, “Intensity Estimation of Spectroscopic Signals with an Improved Sparse Reconstruction Algorithm”, *IEEE Signal Processing Letters*, Vol. 24(5), pp. 530-534, 2017. DOI: 10.1109/LSP.2017.2680839 (Q1, IF 3.2 GR h-5 92) - 9 citations.
- [17] **T. Trigano**, Y. Sepulcre and Y. Ritov, “Sparse Reconstruction Algorithm for Nonhomogeneous Counting Rate Estimation”, *IEEE Transactions in Signal Processing*, Vol. 65(2), pp 372-385, 2017. DOI: 10.1109/TSP.2016.2620104 (Q1, IF 4.6, GR h-5 147) - 13 citations.
- [18] **T. Trigano**, I. Gildin and Y. Sepulcre, “Pileup Correction Algorithm Using an Iterated Sparse Reconstruction Method”, *IEEE Signal Processing Letters*, Vol. 22(9), pp. 1392-1396, 2015. DOI: 10.1109/LSP.2015.2406911 (Q1, IF 3.2 GR h-5 92) - 16 citations.
- [19] D. Luengo, S. Monzon, **T. Trigano**, J. Via and A. Artes-Rodrigues, “Blind Analysis of Atrial Fibrillation Electrograms: Sparsity-Aware Formulation”, *Integrated Computer Aided Engineering*, Vol. 22-1, pp.71-85, 2015. DOI: <https://doi.org/10.3233/ICA-140471> (Q1, IF 5.8, GR h-5 45) - 16 citations.
- [20] **T. Trigano**, E. Barat, T. Dautremer and T. Montagu, “Fast Digital Filtering of Spectrometric Data for Pile-Up Correction”, *IEEE Signal Processing Letters*, Vol. 22(7), pp. 973-977, July. 2014. DOI: 10.1109/LSP.2014.2377352 (Q1, IF 3.2 GR h-5 92) - 12 citations.
- [21] Y. Sepulcre, **T. Trigano** and Y. Ritov, “Sparse Regression Algorithm for Counting Rate Estimation in Gamma Spectrometry”, *IEEE Transactions in Signal Processing*, Vol 61(17), 4347-4359, September, 2013. DOI: 10.1109/TSP.2013.2264811 (Q1, IF 4.6, GR h-5 147) - 31 citations.

Papers published after last promotion

Papers published before last promotion

- [22] R. Goot, **T. Trigano**, S. Tapuchi and J. Gavan, “Adaptative Allocation of Power Transmission for HAPs”, in *Annals of Telecommunications*, Vol 68, 107-113, June 2013. DOI: <https://doi.org/10.1007/s12243-012-0303-1> (Q2, IF 1.8, GR h-5 45) - 3 citations.

- [23] J. Gavan, **T. Trigano**, S. Tapuchi and A. Kuperman, “Mitigation of Mobile Radio Parasitic Radiations Effect: a Review”, in *Journal of Communication Engineering Systems*, Vol 2(1), 1-11 2012, Q4.
- [24] **T. Trigano**, U. Isserles and Y. Ritov, “Semiparametric curve alignment and shift density estimation for biological data”, *IEEE Transactions in Signal Processing*, Vol 59(5), 1970-1984, May, 2011. DOI: 10.1109/TSP.2011.2113179 (Q1, IF 4.6, GR h-5 147) - 27 citations.
- [25] **T. Trigano**, T. Montagu, E. Moulines, F. Roueff and A. Souloumiac, “Statistical Pile-Up Correction Method for HPGe Detectors”, *IEEE Transactions in Signal Processing*, Vol 55(10), 4871-4881, 2007. DOI: 10.1109/TSP.2007.896300 (Q1, IF 4.6, GR h-5 147) - 25 citations.
- [26] E. Moulines, F. Roueff, A. Souloumiac and **T. Trigano**, 2007, “Nonparametric Inference about Photon Energy from Indirect Measurements”, *Bernoulli*13(2), 365-388, 2007, 10.3150/07-BEJ5184 (Q1, IF 1.7, GR h-5 76) - 14 citations.

Contributions in the publications:

- In all the papers on which I am the first author beside [5], and in [26], I performed all the numerical experiments, wrote more than 50% of the paper and contributed to the theoretical results.
- In [3] and [2], I contributed to the algorithmic development, experiments design and the paper’s editing.
- In [21], [12] and [5], I contributed equally on the theoretical results, numerical experiments and writing.
- In [4], [9] and [14] I performed the numerical analysis and wrote the associated section of the paper.
- In , [19], [22], [23], I was responsible of the ML/algorithmic parts and the experiments. I also wrote a significant part of the paper.
- In [1], I contributed to the paper writing and editing.

Accepted for publications:

N/A

E. Articles or Chapters in Scientific Books (which are not Conference Proceedings)

- [1] Z. Chen, X. Kong, X. Zheng, Y. Zhu and **T. Trigano**, “Parallel pileup correction for nuclear spectrometric data on many-core accelerators”, in *Smart Computing and Communications*, M. Liu, Z. Liu and C. Zhang (eds), Lecture Notes in Computer Science, vol. 13828, pp 258-267, Springer, 2022. DOI: https://doi.org/10.1007/978-3-031-28124-2_24 .

- [2] D. Meltzer, D. Luengo and **T. Trigano**, “ECG-based Biometric Recognition”, in *Teaching and Subjects on Biomedical Engineering*, pp 157-171, Acco CV, Leuven, Belgium, 2021
- [3] D. Luengo, D. Oses and **T. Trigano**, “Digital Signal Processing of ECG and PCG signals”, in *Teaching and Subjects on Biomedical Engineering*, pp 132-156, Acco CV, Leuven, Belgium, 2021 - 1 citation.
- [4] D. Luengo, D. Meltzer and **T. Trigano**, “Overcomplete Multiscale Dictionaries for Efficient Representations of ECG Signals”, in *Computer Aided Systems Theory*, Moreno-Díaz, R., Pichler, F., Quesada-Arencibia, A. (eds), Lecture Notes in Computer Science, vol. 12014, pp 331-338, 2020. DOI: https://doi.org/10.1007/978-3-030-45096-0_41 - 1 citation.
- [5] **T. Trigano** and Y. Sepulcre, “Regularized Sparse Representation for Spectrometric Pulse Separation and Counting Rate Estimation”, in *Latent Variable Analysis and Source Separation*, Theis, F., Cichocki, A., Yeredor, A., Zibulevsky, M. (eds), Lecture Notes in Computer Science, Vol 7191, 188-195, 2012. DOI: https://doi.org/10.1007/978-3-642-28551-6_24.

Chapters published after last promotion

Chapters published before last promotion

- [6] T. Trigano, U. Isserles, T. Montagu and Y. Ritov, “Semiparametric curve alignment and shift density estimation : ECG data processing revisited”, in *Signal Processing*, S. Miron (Ed), InTechOpen, pp. 217-240, 2010, <https://doi.org/10.5772/8526> (2 citations)

Contributions in the publications:

- In all the contributions on which I am the first author, I performed all the numerical experiments, wrote more than 50% of the paper and contributed to the theoretical results.
- In [1-4], the authors contributed equally.

F. Articles in Conference Proceedings

All the conference papers went under a review process of two to five reviewers. Most of them underwent correction requirements. In all the papers below my contributions were significant in all aspects.

- [1] D. Luengo, A. Treytl, P. Arras, K. Kornijenko, G. Tabunshchyk and **T. Trigano**, “Improving Quality of Life through the Engineering Education. Case Study”, *Proceedings of E-TEMS*, 9-11 March 2022, Bilbao, Spain, pp 190-195, 2022. DOI: 10.1109/E-TEMS53558.2022.9944453 - 2 citations.

- [2] D. Luengo, J. Via and **T. Trigano**, “Efficient Iteratively Rewighted LASSO Algorithm for Cross-Products Penalized Sparse Solutions”, *Proceedings of EURASIP Conference in Signal Processing (EUSIPCO, Qualis rank: B1)*, 18-21 January 2021, Amsterdam, Netherlands, 5 pages, 2020. DOI: 10.23919/Eusipco47968.2020.9287804 - 4 citations.
- [3] D. Luengo, D. Meltzer and **T. Trigano**, “Sparse ECG Representation with a Multi-Scale Dictionary Derived from Real-World Signals”, *41th International Conference on Telecommunications and Signal Processing*, 04-06 July 2018, Athens, Greece, 4 pages, 2018. DOI: 0.1109/TSP.2018.8441329 - 8 citations
- [4] **T. Trigano**, V. Kolesnikov, D. Luengo Garcia and A. Artes-Rodrigues, “Grouped Sparsity Algorithm for Multichannel Intracardiac ECG Synchronization”, *Proceedings of EURASIP Conference in Signal Processing (EUSIPCO, Qualis rank: B1)*, 01-05 September 2014, Lisbon, Portugal, 2014. Link: <https://ieeexplore.ieee.org/document/6952547> - 12 citations.
- [5] D. Luengo, J. Via, S. Monzon, **T. Trigano** and A. Artes-Rodrigues, “Cross-Products LASSO”, in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP, Qualis rank: A1)*, 26-31 May 2013, Vancouver, Canada, 5 pages, 2013. DOI: 10.1109/ICASSP.2013.6638840 - 18 citations.
- [6] M. Lopatin, N. Moskovitch, **T. Trigano** and Y. Sepulcre, “Pileup Attenuation for Spectroscopic Signals Using A Sparse Reconstruction”, in *Proceedings of 27-th IEEE Convention of Electrical and Electronics Engineers in Israel*, 14-17 November 2012, Eilat, Israel, 5 pages, 2012. DOI: 10.1109/EEEI.2012.6377045 - 7 citations.
- [7] S. Monzon, D. Luengo Garcia, **T. Trigano** and A. Artes-Rodrigues, “Sparse Spectral Analysis Of Atrial Fibrillation Electrograms”, in *2012 IEEE International Workshop on Machine Learning for Signal Processing*, 23-26 September 2012, Santander, Spain, 5 pages, 2012. DOI: 10.1109/MLSP.2012.6349721 - 23 citations.

Papers published after last promotion

Papers published before last promotion

- [8] Y. Sepulcre and **T. Trigano**, “Iterated Sparse Reconstruction for Activity Estimation in Nuclear Spectroscopy“, in *Proceedings of EURASIP Conference in Signal Processing (EUSIPCO, Qualis rank: B1)*, 27-31 August 2012, Bucharest, Romania, 5 pages, 2012, <https://ieeexplore.ieee.org/document/6334066> - 1 citation.
- [9] **T. Trigano**, Y. Sepulcre, M. Roitman and U. Aferiat, “On Nonhomogeneous Activity Estimation in Gamma Spectrometry using Sparse Signal Representation”, in *2011 IEEE Statistical Signal Processing Workshop (SSP)*, 28-30 June 2011, Nice, France, 649-652, 2011. DOI: 10.1109/SSP.2011.5967784 - 13 citations.
- [10] **T. Trigano**, Y. Sepulcre, M. Tal and Y. Mashiach, “Sparse Regression Algorithm for Activity Estimation in Gamma Spectrometry”, in *Proceedings of 26th IEEE Convention of Electrical and Electronics Engineers in Israel*, 17-20 November 2010, Eilat, Israel, 268-272, 2010. DOI: 10.1109/EEEI.2010.5661973 - 3 citations.

- [11] **T. Trigano**, U. Isserles and Y. Ritov, “Semiparametric Density Estimation of Shifts for alignment of ECG data”, in *EURASIP Signal Processing Conference (EUSIPCO, Qualis rank: B1)*, August 25-29 2008, Lausanne, Switzerland, 1-5, 2008, <https://www.eurasip.org/Proceedings/Eusipco/Eusipco2008/papers/1569101580.pdf> - 3 citations.
- [12] T. Dautremer, E. Barat and **T. Trigano**, “Nonparametric Bayesian Estimation of Censored counters Intensity from the indicator data”, in *AIP Conference Proceedings, Vol 872(485), International Workshop on Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, Paris, France, 8 pages, 2006. DOI: 10.1063/1.2423310 - 2 citations.
- [13] **T. Trigano**, F. Roueff, E. Moulines, A. Souloumiac and T. Montagu, “Energy Spectrum Reconstruction for HPGe Detectors Using Analytical Pile-Up Correction”, in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP, Qualis rank: A1)*, 14-19 May 2006, Toulouse, France, Volume 3, III-VII, 2006. DOI: 10.1109/ICASSP.2006.1660723 - 2 citations.
- [14] **T. Trigano**, F. Roueff, A. Souloumiac and E. Moulines, “Nonparametric Inference for Pile-up Correction in Nuclear Spectrometry”, in *IEEE Statistical Signal Processing Workshop (SSP)*, 17-20 July 2005, Bordeaux, France, 754-750, 2005. DOI: 10.1109/SSP.2005.1628693 - 3 citations.
- [15] **T. Trigano**, E. Barat, T. Dautremer and A. Souloumiac, “Pile-up Correction Algorithms for Nuclear Spectrometry”, in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP, Qualis rank: A1)*, 23-23 March 2005, Philadelphia, USA, iv/441 – iv/444, 2005. DOI: 10.1109/ICASSP.2005.1416040 - 18 citations.
- [16] **T. Trigano**, D. Mostefa and G. Chollet, “Implementation of a classification tree in a ASV system”, *Proceedings of the COST-275 Workshop*, Paris, 6 pages, 2002.
- [17] **T. Trigano**, D. Mostefa, G. Chollet and D. Petrovska, “Description of the ENST Speaker Verification System”, *NIST Speaker Recognition Workshop*, Washington, 5 pages, 2002.

Accepted for publications:

N/A

G. Other Scientific Publications

Patents

- [1] L. Schwartz, **T. Trigano**, Y. Bechor. “Process to isolate object of interest in image”, US Patent US10417772B2, <https://patents.google.com/patent/US10417772B2/en>, 2019 - 8 citations.

- [2] E. Barat, T. Dautremer, **T. Trigano** and T. Brisset, “Measurement and treatment of a signal comprising stacks of elementary pulses”, US patent 11579757, 40 pages, 2008 - 7 citations.
- [3] E. Barat, T. Dautremer, **T. Trigano** and T. Brisset, “System and apparatus of processing of a signal consisting in pileups of individual pulses”, European patent 2005773039, 49 pages, 2007 - 5 citations.

Technical reports:

- [1] **T. Trigano**, “Deconvolution methods for discrete histograms: multiplicative gradient algorithm for denoising energy spectra”, CEA Saclay, Technical Report 05-031, 29 pages, 2005.
- [2] **T. Trigano**, “Numerical inversion of the Laplace transform: application for pileup correction of continuous-time signals for spectrometry”, CEA Saclay, Technical Report 04-015, 44 pages, 2004.
- [3] E. Barat, T. Dautremer and **T. Trigano**, “Stochastic modeling for the photonic signal: pileup correction algorithm for discrete-time signals from spectrometry”, CEA Saclay, Technical Report 03-018, 44 pages, 2003.

H. Other Publications

- [1] **T. Trigano** and Dima Bykhovsky, “Digital Signal Processing”, course booklet for “Digital Signal Processing” taught at SCE, 120 pages, 2023 (Hebrew).
- [2] **T. Trigano**, “Digital Image Processing”, course booklet for “Medical Image Processing 1” taught at Tel-Aviv University, 132 pages, 2022 (English).
- [3] **T. Trigano**, “Mathematical Methods Using Python”, course booklet for “Numerical Methods with Python” taught at SCE, 84 pages, 2019 (English).
- [4] **T. Trigano**, “Introduction to random processes”, course booklet for “Random Signals” taught at Afeka, 81 pages, 2018 (Hebrew).
- [5] **T. Trigano**, “Information Theory”, course booklet for “Information Theory” taught at SCE, 98 pages, 2017 (Hebrew).
- [6] **T. Trigano**, “Signals and Systems”, course booklet for “Signals and Systems” taught at SCE, 98 pages, 2016 (Hebrew).
- [7] **T. Trigano**, O. Ben Harush and M. Nanikashvili “Introduction to Probability and Statistics”, course booklet for “Probability and Statistics” taught at SCE, 134 pages, 2011 (Hebrew).

I. Submitted Publications

Submitted Journal Publications:

- [1] A. Mishra, A. Krief, M.M. Sahoo, A. Schachter, I. Gondac, N. Dudaic, **T. Trigano**, I. Herrmann, "Pre- and post-harvest spectral estimation of selected phytochemicals in rosemary", submitted to Computers and Electronics in Agriculture, 2024.
- [2] Z. Chen, X. Zheng, Y. Zhu, **T. Trigano**, Y. Huang, and Y. Zhang, "Spectral Pileup Correction and Noise Reduction Using CUDA Multi-Core Parallel Processing", submitted to Nuclear Science and Techniques, 2024.
- [3] Y. Sepulcre, S. Zaretsky and **T. Trigano**, "C-LASSO: Adding Convolutions in LASSO for Sparser Solutions in Convolutional Dictionaries", submitted to IEEE Transactions in Signal Processing, 2024.